CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 8: **Introduction to Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a right triangle with perpendicular 4, base 3 and hypotenuse 5, $\sin\theta =$ 1

(a) \frac{5}{4} **(b)** \frac{4}{5} **(c)** \frac{4}{3} **(d)** \frac{3}{5}

2. In a right triangle with perpendicular 4, base 3 and hypotenuse 5, $\tan\theta =$ 1

(a) \frac{4}{5} **(b)** \frac{4}{3} **(c)** \frac{3}{5} **(d)** \frac{3}{4}

3. The value of $sin30^\circ$ is: 1

(a) \frac{1}{2} **(b)** 1 **(c)** 0 **(d)** \frac12

4. The value of $\sin^2 20^\circ + \cos^2 20^\circ$ is: 1

(a) 0 **(b)** 2 **(c)** 1 **(d)** 20

5. $\sin 25^\circ = \cos$ of: 1

(a) 25^\circ **(b)** 40^\circ **(c)** 65^\circ **(d)** 155^\circ

6. $\sec\theta$ is the reciprocal of: 1

(a) $\tan\theta$ **(b)** $\cos\theta$ **(c)** $\sin\theta$ **(d)** $\cot\theta$

7. In a right triangle with perpendicular 12, base 5 and hypotenuse 13, $\sin\theta =$ 1

(a) \frac{13}{12} **(b)** \frac{12}{5} **(c)** \frac{5}{13} **(d)** \frac{12}{13}

8. In a right triangle with perpendicular 12, base 5 and hypotenuse 13, $\tan\theta =$ 1

(a) \frac{12}{5} **(b)** \frac{5}{12} **(c)** \frac{5}{13} **(d)** \frac{12}{13}

9. The value of $cos60^\circ$ is: 1

(a) \frac{1}{2} **(b)** 0 **(c)** 1 **(d)** \frac12

10. The value of $\sin^2 35^\circ + \cos^2 35^\circ$ is: 1

(a) 1 **(b)** 2 **(c)** 0 **(d)** 35

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In a right triangle the perpendicular is 4 and the hypotenuse is 5. Find $\sin\theta$. 2

12. Find the value of $sin30^\circ + cos60^\circ$. 2

13. Evaluate: $\sin^2 25^\circ + \cos^2 25^\circ$. 2

14. In a right triangle the perpendicular is 12 and the hypotenuse is 13. Find $\sin\theta$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, right-angled at B, $AB = 3$ cm and $BC = 4$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

16. If $\sin A = \frac{4}{5}$, find $\cos A$ and $\tan A$. 3

17. Evaluate: $\sin^2 30^\circ + \cos^2 60^\circ$. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 5$ cm and $BC = 12$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

18. If $\sin A = \frac{12}{13}$, find $\cos A$ and $\tan A$. 3

OR

Evaluate: $\tan^2 45^\circ + \sin^2 90^\circ$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. If $\tan \theta = \frac{4}{3}$, find $\sin \theta$, $\cos \theta$ and $\sec \theta$. 5

OR

In $\triangle PQR$, right-angled at Q, $PQ = 6$ cm and $QR = 8$ cm. Find $\sin P$, $\cos P$, $\tan P$ and check that $\sin^2 P + \cos^2 P = 1$. 5

20. If $\tan \theta = \frac{12}{5}$, find $\sin \theta$, $\cos \theta$ and $\sec \theta$. 5

OR

In $\triangle PQR$, right-angled at Q, $PQ = 15$ cm and $QR = 36$ cm. Find $\sin P$, $\cos P$, $\tan P$ and check that $\sin^2 P + \cos^2 P = 1$. 5

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND